# 10 kW Orbital Node

*Public Mission Definition and Pre-SRR Baseline*

**Compute where space data begins.**

*Notional concept render. This is not flight CAD and does not establish interfaces, dimensions or launch compatibility.*

> A first owned LEO system delivering 10 kW of continuous electrical input to the compute payload. This package defines the mission and verification intent; it is not flight CAD, an ICD or a launch-provider compatibility statement.

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PRE-SRR BASELINE

## 1. Mission statement

**TOP-LEVEL REQUIREMENT**

The mission shall demonstrate customer-relevant processing of space-originated data while delivering 10 kW continuous electrical power at the compute-payload input during nominal operations, including the design eclipse where applicable.

| 10 kW | 500-600 km | 3-7 t |
|---|---|---|
| PAYLOAD INPUT | LEO TRADE SPACE | WET-MASS ASSUMPTION |

| Parameter | Public baseline | Status |
|---|---|---|
| Mission class | First owned orbital processing node | REQUIREMENT |
| Orbit | 550 km circular LEO screening case; SSO local time TBD | WORKING ASSUMPTION |
| Nominal mission life | 5 years; 7-year design sensitivity | WORKING ASSUMPTION |
| Compute payload | 10.0 kW continuous at DC input | REQUIREMENT |
| Full spacecraft load | 12-15 kW; nominal 13.5 kW | WORKING ASSUMPTION |
| Wet launch mass | 3-7 tonnes; nominal planning point 4.7 tonnes | WORKING ASSUMPTION |
| Launch class | Heavy rideshare or dedicated | TBD BY SUPPLIER |
| Initial customer workflow | SAR maritime vessel detection and scene prioritization | BUSINESS HYPOTHESIS |

WHAT THE FLIGHT MUST PROVE

# 2. Mission objectives

| ID | Objective | Success evidence |
|---|---|---|
| MO-01 | Commission and maintain positive spacecraft control | Healthy power, attitude, communications and thermal telemetry |
| MO-02 | Operate the compute system at 1, 3-5 and 10 kW levels | Stable staged ramp with no limit violations |
| MO-03 | Sustain 10 kW continuous payload input | Agreed-duration run through the design environmental case |
| MO-04 | Process a customer-defined space-originated dataset | Accepted quality, latency and data-reduction results |
| MO-05 | Demonstrate fault containment and recovery | Injected hardware, software and link failures recover within limits |
| MO-06 | Demonstrate secure workload and data lifecycle | Signed load, provenance, encryption, access and deletion evidence |
| MO-07 | Preserve end-of-life disposal and passivation | Verified maneuver, energy and command reserve |

## Non-objectives

- Proving commodity cloud economics or serving arbitrary public workloads.
- Operating in GEO or at a Lagrange point.
- Manufacturing every sensor, spacecraft subsystem or launch vehicle.
- Claiming 10 kW of accelerator-silicon consumption.
- Scaling to 100 kW before first-node customer and operations evidence exists.

SENSOR TO DECISION

# 3. Concept of operations

| Phase | Activities | Critical constraint |
|---|---|---|
| Ground preparation | Customer workload, data rights, model, keys and acceptance plan are frozen | No uncontrolled model or data reaches flight |
| Launch and early orbit | Deploy, establish control, inspect power and thermal state | Compute remains disabled until platform health is proven |
| Compute commissioning | Ramp one tile, then multiple tiles, then full cluster | Each step has automatic abort limits |
| Service operation | Receive sensor data, process, store, and downlink or crosslink results | Data path and customer priority are scheduled |
| Contingency | Throttle, isolate tile, reboot domain, enter safe mode | TT&C and disposal capability are protected |
| End of life | Deorbit, passivate and close data custody | Regulatory and customer obligations remain enforceable |

## Data-path alternatives

| Architecture | Advantage | Primary risk |
|---|---|---|
| Same-platform sensor | Simpler ingress and timing | Sensor integration can dominate spacecraft configuration |
| Optical crosslink from partner spacecraft | Shared node can serve multiple operators | Acquisition geometry, terminal interoperability and cost |
| High-rate RF crosslink | Potentially broader partner compatibility | Spectrum, antenna size, interference and licensing |
| Store-and-forward upload | Useful for software and limited test data | Does not establish the core space-originated-data advantage |

LEVEL-1 AND LEVEL-2 INTENT

# 4. System requirements baseline

| ID | Requirement | Verification |
|---|---|---|
| SYS-001 | The system shall deliver 10.0 kW continuous at the compute-payload DC interface in nominal operation. | Test + analysis |
| SYS-002 | The system shall protect TT&C, attitude control and disposal from compute-domain failures. | Test + analysis |
| SYS-003 | The system shall measure power, temperature, flow and error state at tile and domain level. | Inspection + test |
| SYS-004 | The system shall isolate and restart a failed compute tile without full-node loss. | Fault-injection test |
| SYS-005 | The system shall maintain a cryptographically controlled workload and data chain. | Inspection + demonstration |
| SYS-006 | The system shall process at least one customer-defined workload to agreed acceptance metrics. | Demonstration |
| SYS-007 | The system shall support a safe throttled mode when power, thermal or link constraints are exceeded. | Test |
| SYS-008 | The system shall retain deorbit and passivation capability after credible single failures. | Analysis + test |
| SYS-009 | The system shall expose versioned electrical, mechanical, thermal, data and operational interfaces. | Inspection |

An Interface Control Document, or ICD, will define the agreed boundary between two parties or subsystems: voltages, connectors, data protocols, thermal rejection, geometry, loads, responsibilities and verification. This mission definition precedes the ICD; it does not substitute for one.

BETA-ZERO 550 KM SCREEN

## 5. Current sizing envelope

| Metric | Planning range | Nominal | Evidence class |
|---|---|---|---|
| Full spacecraft load | 12-15 kW | 13.5 kW | WORKING ASSUMPTION |
| Solar-array BOL | ~30-37 kW | ~34 kW | CALCULATED |
| Active-PV equivalent | ~100-124 m2 | ~112 m2 | CALCULATED |
| Gross solar planform | ~124-156 m2 | ~140 m2 | CALCULATED |
| Battery nameplate | ~25-31 kWh | ~28 kWh | CALCULATED |
| Heat rejection | ~12-15 kW | ~13 kW | CALCULATED |
| Equivalent radiator area | ~33-42 m2 | ~37 m2 | CALCULATED |
| Physical radiator planform | 35-60 m2 | TBD | WORKING ASSUMPTION |
| Wet launch mass | 3-7 tonnes | 4.7 tonnes | WORKING ASSUMPTION |
| Approximate deployed span | 25-40 m | TBD | WORKING ASSUMPTION |

The calculated screen is not a subsystem allocation. It omits detailed incidence, temperature, harness, converter, structural, deployment, contamination, attitude, shielding and propellant effects. Supplier-backed budgets must replace the planning ranges by PDR.

> **SUPPLIER BOUNDARY**
>
> Current public bus claims are leads, not closure. Apex's Comet Mini page lists 8 kW orbit-average payload power and 6 kW continuous dissipation. These vendor-published figures do not establish compatibility with this mission's 10 kW continuous electrical input at the compute-payload interface or its heat-rejection requirements. EnduroSat advertises scalable generation and a multi-tonne payload class, but its public values do not prove a turnkey 10 kW compute-payload electrical service.

PRELIMINARY OWNERSHIP

# 6. Subsystem allocation and interfaces

| Subsystem | Preliminary responsibility | Interface to freeze by PDR |
|---|---|---|
| Compute payload | Satellite Inference | DC power, heat pickup, data, timing, telemetry and mechanical envelope |
| Bus and primary structure | Bus supplier / co-development | Mass properties, stiffness, loads, volume and safe-mode services |
| Solar and PMAD | Array and EPS suppliers | BOL/EOL power, voltage, MPPT, pointing, deployment and protection |
| Battery | EPS/battery supplier | Energy, voltage, cycle life, DoD, thermal and fault containment |
| Thermal transport and radiator | Co-developed with thermal supplier | Heat load, temperatures, flow, redundancy, view factor and deployment |
| ADCS and propulsion | Bus supplier | Flexible-body modes, momentum, pointing, maneuver and disposal |
| Communications | Bus/comms suppliers plus Satellite Inference | RF/optical links, data rates, acquisition, encryption and scheduling |
| Ground and mission operations | Satellite Inference plus service providers | Command authority, telemetry, security, customer API and incident response |
| Launch integration | Launch integrator and provider | Envelope, loads, coupled loads, separation and range documentation |

WHAT CAN BE PROVEN BEFORE FLIGHT

# 7. Ground development and verification matrix

| Test article | Test | Pass criterion | Flight risk reduced |
|---|---|---|---|
| 1 kW tile | 1,000-hour representative workload | No unplanned reset; stable power, temperature and throughput | Compute and local cooling |
| 1 kW tile | Fault injection | Errors contained; watchdog and restart meet limit | Radiation-like upset response |
| 1 kW tile | Power transient | No unsafe bus interaction or data corruption | EPS interface |
| 10 kW breadboard | Full-load steady state | 10 kW payload input with closed heat balance | System thermal transport |
| 10 kW breadboard | N-1 tile and pump case | Graceful degradation and protected platform loads | Fault tolerance |
| Network rig | Sensor-to-result flow | Customer p95 latency, accuracy and data reduction met | Commercial mission closure |
| Engineering model | EMI/EMC and interfaces | Margins against allocated limits | Integration |
| Structural article | Modal, vibration and deployment | Correlated model and positive margins | Launch and deployment |
| Thermal-vacuum article | Hot/cold and transitions | No limit violation; correlated thermal model | Orbital environment |
| Flight system | End-to-end mission rehearsal | Nominal and contingency procedures accepted | Operations |

## Software acceptance metrics

- p95 sensor-to-decision latency under defined data volume and link conditions.
- Recall at the customer's maximum acceptable false-alarm rate.
- Accepted outputs per kWh and raw-to-result data-reduction ratio.
- Availability by workload class and recovery time by failure class.
- Cryptographic identity, provenance, deletion and audit evidence.

NO LAUNCH BOOKING BEFORE GEOMETRY

## 8. Reviews, schedule logic and decision gates

| Gate | Minimum package | Go decision |
|---|---|---|
| MDR | Mission need, workload and data path, LEO trade, first customer hypothesis | Mission is coherent enough for requirements |
| SRR | Requirements, tile test, initial suppliers, compliance plan and V&V map | Requirements are necessary and testable |
| PDR | 10 kW breadboard, selected architecture, budgets, CAD concept, ROMs and customer evidence | Architecture can meet requirements within margin |
| CDR | Released design, analyses, qualification procedures, ICDs and launch interface | Design is ready to build and qualify |
| FRR | Qualified hardware, licenses, operations, range and launch readiness | Risk is acceptable for launch |

Launch price cannot be meaningfully requested as a binding quote until mass, stowed envelope, separation system, orbit, schedule, hazardous materials and integration responsibilities are known. Early launch-provider discussions are still useful for planning envelopes, compatible dispensers, lead time and documentation expectations.

MDR CLOSURE LIST

# 9. Open items and required decisions

| Decision | Options | Required evidence |
|---|---|---|
| Same-platform sensor or crosslinked node | One sensor, multiple sensors, hybrid | Customer and partner data-path study |
| Ordinary SSO or high-beta trade | Mid-morning/afternoon, dawn-dusk sensitivity | Lighting, beta, thermal and launch availability |
| Single spacecraft or distributed pair | One 10 kW node, two 5 kW nodes | Mass, reliability, network and cost trade |
| Compute hardware | Accelerator, CPU, memory, storage and redundancy mix | Customer benchmarks plus radiation screen |
| Thermal architecture | Pumped loop, heat pipes, cold plates, deployable radiators | Thermal model and supplier study |
| Launch class | Heavy rideshare or dedicated | Supplier-backed mass properties and envelope |
| Operational customer | Defense/sovereign, EO partner, maritime or disaster program | Paid evaluation and conditional reservation |

## Sources

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[10] U.S. Bureau of Industry and Security, export administration regulations. https://www.bis.gov/regulations/ear

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